Infineon/BIT Excellence Program 2026

Sustainable Supply Chain for Dried Mango from Africa to Germany

Module 08
Machinery Management System

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1 Machinery Management Large Scope Definition and Equipment Registry

1.1 Scope and Perimeter of Work

The Machinery Management System is designed to track and manage every piece of mechanized or hand-operated equipment used across the farm's production chain, from mango cultivation to the processing of dried mango. Rather than treating equipment as a side note within each production stage, this module gathers all physical assets into a single registry so that their maintenance, usage, and lifecycle can be monitored independently of the process they support.

In scope are all equipment items directly involved in growing, harvesting, and processing mangoes: agricultural machinery such as the tractor or power tiller and the motor pump used for irrigation, hand tools used for planting and harvesting, post harvest handling equipment, and the full processing line made up of the elevators, washer, sorter, cleaner, dryer, and peeler. For each of these assets, the system covers six areas: maintenance scheduling, usage and operating hours, repair history, fuel consumption, operator assignment, and spare parts inventory.

Out of scope, for now, are the farm's energy generation assets, namely the solar panels and batteries covered under the Energy Supply Systems module, and the vehicles used to transport dried mango from Banfora to Germany, which are assumed to fall under the Sales and Marketing module's delivery tracking. These two boundaries still need to be confirmed with the owners of the modules concerned.

A deliberate distinction is made between owning a process and owning an asset. The Water Supply module remains responsible for irrigation scheduling and water quality, and the Product Transformation module remains responsible for drying quality and food safety compliance. The Machinery module, by contrast, is only responsible for the physical equipment itself, such as the motor pump or the dryer, and for keeping that equipment operational. This separation avoids duplicating data between modules while still giving Machinery full visibility over every asset the farm depends on.

1.2 Data Consolidation from 2025 Cohort

Before defining this year's equipment list, the 2025 cohort's Excel file was reviewed sheet by sheet to identify any existing data that could serve as a starting point, in line with the instruction to study last year's sustainable farm and use it as a foundation.

On Zalka's Plants sheet, Table 4 (Equipment Pricing) lists five items: tractor and power tiller, motor pump and irrigation, planting and harvesting tools, post harvest equipment, and inputs such as fertilizers and seeds. The total shown, 15,548 EUR, is the sum of the first four lines only, correctly leaving inputs out of the equipment total. Table 6 (Purchase Price of Plots), however, later lists an equipment line of 19,362 EUR, which corresponds to 15,548 EUR plus the 3,814 EUR of inputs added back in. This inconsistency, inputs being excluded from equipment in one table and included in another, was one of the first issues identified in the source file. It confirms that inputs such as fertilizers and seeds should be treated as operational costs rather than equipment throughout this module.

The same sheet's bibliography section also clarifies what each rolled up line actually contains. The tractor and power tiller line and the motor pump and irrigation line correspond respectively to a mini tractor priced at 6,860 EUR and a motor pump with hose kit priced at 4,573 EUR. The planting and harvesting tools line, at 1,829 EUR, covers a steel hoe, a branch cutter, pruning shears, nursery trays, a manual row transplanter, a wheelbarrow, and a thermal motorized sprayer. The post harvest equipment line, at 2,286 EUR, covers a sorting table, a brushing unit, a belt conveyor, a packing scale, an impulse sealer, a manual pallet jack, and plastic crates. Once these itemized lists were matched against the rolled up totals, none of the original categories turned out to be arbitrary, they had simply been summarized without their supporting detail.

Cedric's Smart Dryer sheet contributed the processing side of the equipment list: elevators, bubble washer, sorter, cleaner, dryer, and peeler, priced individually and totaling 8,861 EUR. These items were kept in a separate sheet from Zalka's list in last year's work, even though they belong to the same physical value chain. This year, they have been merged into a single registry rather than split by production stage.

Abdoul's work on the solar PV power plant was reviewed as well, but the battery and solar tracking data it contains describes energy generation assets rather than mechanized farm equipment. It has therefore been left out of this module and is expected to remain under Energy Supply Systems.

The logic applied to this year's Machinery module follows directly from these observations: every physical, mechanized or hand-operated asset used in cultivation, harvest, or processing is tracked once, under a single asset registry, regardless of which stage of production it supports, while consumables such as fertilizers, seeds, fuel, and packaging materials, along with energy generation assets, are explicitly kept out.

1.3 Consolidated Equipment List

The table below merges the equipment data scattered across last year's Plants and Smart Dryer sheets into a single list, with fictional asset IDs assigned for use in this year's prototype. Unit costs are carried over from the 2025 figures, since they still represent the best available cost estimate for a 2-hectare farm and allow this year's work to build directly on last year's business case.

Asset ID	Equipment	Category	Stage	Source
MCH-001	Tractor / power tiller	Agricultural machinery	Cultivation	Zalka
MCH-002	Motor pump (irrigation asset)	Agricultural machinery	Cultivation	Zalka
MCH-003	Planting & harvesting tool set	Agricultural equipment	Cultivation	Zalka
MCH-004	Post-harvest handling equipment	Agricultural equipment	Harvest	Zalka
MCH-005	Elevators	Processing machinery	Processing	Cedric
MCH-006	Bubble washer	Processing machinery	Processing	Cedric
MCH-007	Sorter	Processing machinery	Processing	Cedric
MCH-008	Cleaner	Processing machinery	Processing	Cedric
MCH-009	Dryer	Processing machinery	Drying	Cedric
MCH-010	Peeler	Processing machinery	Processing	Cedric

Asset IDs follow the pattern MCH-XXX and are entirely fictional at this stage, meant to populate the frontend prototype rather than to reflect real serial numbers or tags. Once the farm is operational, each ID would be replaced by a physical asset tag linked to the equipment's actual purchase record.

1.4 Data Model

The data model below turns the six focus areas of this module, namely maintenance, usage, repairs, fuel, operators, and spare parts, into seven linked tables, all built around the Equipment table as the central reference. Every other table refers back to an asset through its `asset_id`, which keeps the model consistent and avoids repeating equipment details in each log. This structure is what will be used to generate the fictional dataset behind the frontend prototype.

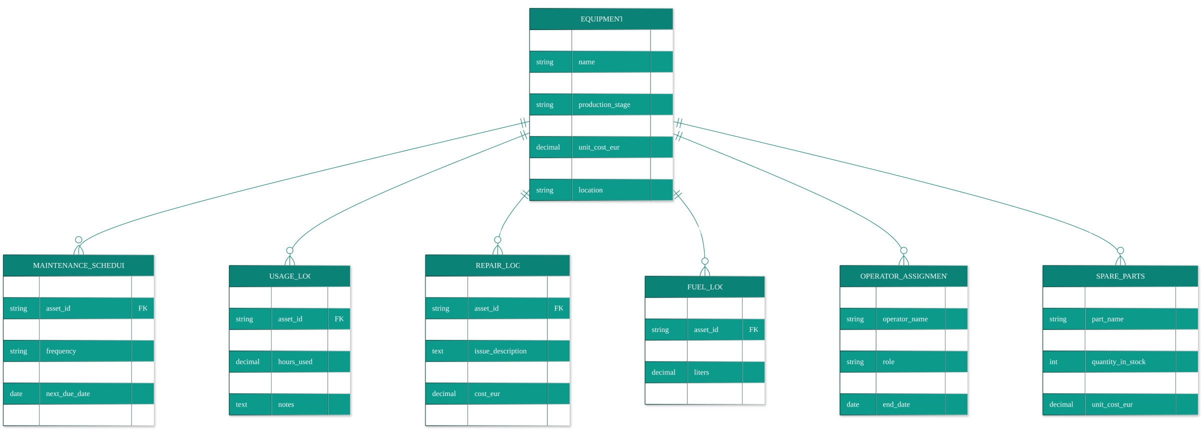

Figure 1: Entity relationship diagram of the Machinery data model.

Equipment

Field	Type	Description
asset_id	string (PK)	Unique identifier, e.g. MCH-001
name	string	Equipment name
category	enum	Agricultural machinery / Agricultural equipment / Processing machinery
production_stage	enum	Cultivation / Harvest / Processing / Drying
purchase_date	date	Date the asset entered service
unit_cost_eur	decimal	Purchase cost in euros
status	enum	Operational / Under maintenance / Out of service
location	string	Site where the asset is currently based

Maintenance Schedule

Field	Type	Description
schedule_id	string (PK)	Unique identifier for the maintenance entry
asset_id	string (FK)	Reference to the Equipment table
maintenance_type	enum	Preventive / Corrective
frequency	string	e.g. every 250 operating hours, every 3 months
last_completed_date	date	Date maintenance was last carried out
next_due_date	date	Date the next maintenance is due
assigned_operator	string (FK)	Reference to the operator responsible

Usage Log

Field	Type	Description
log_id	string (PK)	Unique identifier for the usage entry
asset_id	string (FK)	Reference to the Equipment table
date	date	Date the equipment was used
hours_used	decimal	Operating hours recorded that day
operator_id	string (FK)	Operator who used the equipment
notes	text	Free text field for context

Repair Log

Field	Type	Description
repair_id	string (PK)	Unique identifier for the repair entry
asset_id	string (FK)	Reference to the Equipment table
date	date	Date the repair took place
issue_description	text	Description of the fault
downtime_hours	decimal	Hours the equipment was out of service
cost_eur	decimal	Cost of the repair
technician	string	Person or team who carried out the repair

Fuel Log

Field	Type	Description
fuel_log_id	string (PK)	Unique identifier for the fuel entry
asset_id	string (FK)	Reference to the Equipment table
date	date	Date of refueling
liters	decimal	Quantity of fuel added
cost_eur	decimal	Cost of that refueling

Operator Assignment

Field	Type	Description
assignment_id	string (PK)	Unique identifier for the assignment
operator_name	string	Name of the operator
asset_id	string (FK)	Equipment the operator is assigned to
role	string	e.g. Driver, Technician, Supervisor
start_date	date	Start of the assignment
end_date	date	End of the assignment, if applicable

Spare Parts

Field	Type	Description
part_id	string (PK)	Unique identifier for the spare part
part_name	string	Name of the part
compatible_asset_id	string (FK)	Equipment the part fits
quantity_in_stock	integer	Current stock level
reorder_threshold	integer	Stock level that triggers a reorder alert
unit_cost_eur	decimal	Cost per unit

1.5 Frontend Prototype

A working frontend mockup of the Machinery Management System is available and linked from my personal homepage tracker, where it is kept up to date alongside the rest of the project work. Rather than capturing a static screenshot in this report, readers are invited to check the live version there, as the interface continues to evolve together with the underlying data model.